

MADHUMANTI SAHA

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SUMMARY

Proactive B.Tech 2026 graduate with strong proficiency in **Python** and **JavaScript**, and hands-on experience building full-stack web applications using **React** and **FastAPI**. Skilled in API development, SQL databases, frontend-backend integration, and modern AI tooling (LangChain, vector embeddings, LLMs). Passionate about ownership, clean code practices, and staying updated with AI agents, agentic workflows, and emerging development tools. Eager to contribute as a Software Developer in a fast-paced, impact-driven environment.

EDUCATION

• Academy of Technology, Hoogly <i>Bachelor of Technology in Electrical Engineering (CGPA Obtained: 7.29 (Till 7th Sem))</i>	<i>Hoogly</i> 2022 – 2026
• Udaypur Haradaya Nag Adarsha Vidyalaya, Kolkata <i>Higher Secondary (Percentage Obtained: 77.6%)</i>	<i>Kolkata</i> 2020 – 2022
• Bagbazar Multipurpose Girls' School, Kolkata <i>Secondary (Percentage Obtained: 81.8%)</i>	<i>Kolkata</i> 2015 – 2020

PROJECTS

Personalized Book Recommender System

[GitHub](#)

React — FastAPI — Python — JavaScript — Tailwind CSS — LangChain — ChromaDB — Hugging Face

- Built a full-stack recommendation platform with **React (Vite)** frontend and **FastAPI** backend, implementing RESTful APIs for seamless frontend-backend integration.
- Developed semantic search using **LangChain** + vector embeddings (OpenAI/Hugging Face) and ChromaDB, demonstrating practical LLM integration and agentic AI concepts.
- Implemented hybrid filtering (semantic similarity + metadata) and zero-shot classification for personalized results, showcasing strong problem-solving and data handling skills.
- Ensured responsive UI with Tailwind CSS and clean, maintainable code structure.

Network Security Threat Detection

[GitHub](#)

Python — Scikit-Learn — MLflow — MongoDB — AWS

- Built an end-to-end ML pipeline identifying network threats via log analysis and rule-based anomaly detection (unusual ports, high-frequency requests), enabling faster incident response.
- Performed feature extraction, preprocessing, and model experimentation to improve threat detection accuracy.
- Implemented scalable data processing and training pipelines with MLflow-based experiment tracking.
- Automated model evaluation workflows to support repeatable and reproducible machine learning research.

Customer Churn Analysis & Prediction

[GitHub](#)

SQL — Power Query — DAX — Power BI — Pandas — Scikit-Learn

- Built a complete churn analysis pipeline covering data cleaning, exploratory data analysis (EDA), and feature engineering.
- Developed classification models evaluated using ROC-AUC and F1-score metrics to prioritize customer retention strategies.
- Delivered actionable business visualizations in Power BI, translating model outputs into clear, decision-ready recommendations.

TECHNICAL SKILLS

Programming:	Python, HTML, CSS, JavaScript
Web/APIs:	React, REST APIs, FastAPI, Tailwind CSS
Data:	SQL, NumPy, Pandas
ML/AI:	Scikit-Learn, NLP, Transformers, LangChain
Tools:	Git, GitHub, Docker
Databases:	MySQL, SQL Server, MongoDB
Others:	Power BI, Excel, VS Code, Cursor

ACHIEVEMENTS

- **Salesforce Developer (SmartBridge – AICTE):** Completed 8-week internship; earned Superbadges in Apex, LWC, and Object Relationships. [\[Verify\]](#)
- **Oracle Certified Associate:** MySQL Implementation Certified Associate. [\[Verify\]](#)
- **Smart India Hackathon (SIH) 2024:** Recognized among the Top 8 teams in the national competition.
- **IEEE Student Member:** Active member of IEEE Student Branch (AOT), participating in technical seminars, workshops, and collaborative engineering discussions.